

Fullstack engineer who ships production systems end to end, from data pipelines and services through to the React surfaces on top of them. Currently at Priceline, where I designed and own the conversational memory system behind their AI assistant, built their first ChatGPT and MCP integration, and cut inference spend on a new product surface to a third. Six years across product engineering and applied AI, with a prior career in computational design that shaped how I think about modeling, geometry, and constraints.

Professional Experience

Full Stack Software Engineer (Applied AI) | Priceline.com | 08/2025 - Current | NY

- Designed and shipped Penny Memory, the conversational memory system behind Priceline's AI assistant: Airflow/Composer pipelines synthesizing session data into per customer memory artifacts, served to the assistant at request time. Delivered V1 with one other engineer on the original date; V2 has run in production for six months serving the Penny Canvas product.
- Cut inference spend on the new canvas-studio surface to one third of its prior run rate by introducing prompt caching across the assistant's request paths, with cache keys designed to hold as context grew.
- Built the diagnostic tooling and evaluation criteria for memory quality; measurements showed coverage was low and traced it to skewed input data, redirecting the roadmap. Bounded nightly offline generation to a 200k customer working set after full population runs proved infeasible, holding compute cost flat as usage grew.
- Built Priceline's first ChatGPT integration end to end: the MCP server bridging OpenAI's protocol to internal hotel search services, and the React widget rendered inside ChatGPT's sandbox, making Priceline hotel inventory searchable in natural language.
- Built chat history persistence for images and UI markers, defining the storage path scheme and drawing the responsibility boundary between the persistence service and the backend service so neither took on the other's concerns.
- Made agent behavior debuggable across the org with wrapper utilities emitting OpenInference traces for core agent flows, covering the LangChain, OpenAI Responses, and OpenAI Agents SDK paths.
- Published the architecture docs and ADRs for the memory system and presented it at two engineering roundtables and a company wide dev showcase; separately shipped a Composer DAG re-materializing analytics tables.

Senior Software Engineer | Capital One | 04/2025 - 08/2025 | NY

- Defined and deployed AWS infrastructure as code (CDK, CloudFormation) across Fargate, DynamoDB, ALB and Route53, packaging reusable constructs adopted across the team's services.
- Built and integrated card shipping services into an existing microservice estate, partnering with product on scope and rollout.
- Designed and monitored REST APIs for resiliency and availability at Capital One scale (Java, Python, Splunk, New Relic).

Senior Software Engineer | Outer Labs | 10/2022 - 12/2024 | NY

- Owned the core data model (Postgres, Drizzle ORM) through the pre-seed pivot, and designed the render and state event loop for the space planning web application, the two decisions the rest of the product was built on.
- Led the company's RAG work: ran the experiments and built the backend service that let engineering and product prototype and evaluate LLM workflows themselves (Python, Node.js, Haystack, function calling).
- Built and shipped fullstack prototypes on a weekly cadence while the product thesis was still moving (TypeScript, React, Node.js, Express, Tailwind).
- Renegotiated the API and data model contract between two teams to remove a dependency that was blocking both.
- Wrote the core rendering, state management, and computational geometry libraries the application depends on (Three.js, Canvas API, React).

Software Developer | KOPE | 04/2021 - 10/2022 | London

- Led the development and client engagement to deliver a modular building configurator implementing complex adjacency and stacking rulesets specific to the client's proprietary system
- Contributed to algorithm development and overall implementation during strategic pilot engagements related to building enclosure systems

- Mentored younger engineers on best practices and design patterns

Computational Design Specialist | CallisonRTKL | 02/2020 - 04/2021 | London

- Authored custom plugins for Rhino + Grasshopper relating to the analysis and feasibility of building proposals
- Authored parametric tooling for the analysis of multi-objective tradeoffs of a commercial development
- Implemented a logging system for tracking usage of our internal tooling initiatives across the firm

Education

MSc Architectural Computation | University College London | 09/2018 - 05/2019 | London

Computer-science-for-architects program where we:

- Learned object-oriented-programming, familiarized ourselves with Visual Studio and Git
- Leveraged Unity and C# as a prototyping and simulation environment
- Implemented vanilla L-Systems, Genetic Algorithms, Kohonen Nets, Artificial Neural Networks, Octrees

Bachelor's of Architecture | University of Arizona 08/2008 - 05/2013 | Tucson

Skills

LANGUAGES: TypeScript, JavaScript, Python, C#, SQL, Java, Go

FRONTEND: React, Next.js, Tailwind, MUI, Three.js / R3F, Canvas API, Mapbox, PWA

BACKEND & DATA: Node.js, Express, NestJS, Flask, REST, Postgres, DynamoDB, Firestore, Drizzle, TypeORM, Airflow / Cloud Composer, DuckDB, ETL, BigQuery

CLOUD & INFRA: GCP, Docker, GitHub Actions, GitLab CI, Splunk, New Relic

AI ENGINEERING: MCP, RAG, function calling, prompt caching, LangChain / LangGraph, OpenAI Agents SDK, Haystack, LlamaIndex, Google ADK, OpenInference tracing, evals

ALSO: Rhino, Grasshopper, Revit, Dynamo, Unity, PyTorch, optimization algorithms

Projects

[browse-nyc \(2026\)](#) Event discovery PWA built with React, Vite, TypeScript, Mapbox and Firestore, plus a fleet of Python scrapers feeding the event index.

[NYC-Monitor \(2025\)](#) Multi-agent civic monitoring system, Google ADK competition entry.

[Untabbed \(2024\)](#) Gemini API Developer Competition entry.

[PlanBee \(2020\)](#) Cell-based spatial analysis plugin for Grasshopper, cited in Rumor 76.